Serotonin induces a DOWN state across the anesthetized mouse brain

Rosa Großmann[1], Jorge F. Mejias[2], International Brain Laboratory, Zachary F. Mainen[3] & Guido T. Meijer[3,4]

[1]*Max Planck Institute for Human Cognitive and Brain Science, Leipzig, Germany*
[2] *Cognitive and Systems Neuroscience Group, Swammerdam Institute for Life Sciences, University of Amsterdam, Amsterdam, The Netherlands*
[3]*Champalimaud Foundation, Lisbon, Portugal*
[4]*Donders Centre for Neuroscience, Radboud University Nijmegen, The Netherlands*

Serotonergic neurons in the dorsal raphe nucleus (DRN) project extensively throughout the forebrain, yet their influence on global network states remains controversial. Serotonin is implicated in regulating slow-oscillations, so-called UP and DOWN states, which occur during sleep and light anesthesia. While optogenetic fMRI suggests that serotonin (5-HT) suppresses brain-wide activity, classical electrical stimulation studies report the induction of cortical UP states. To resolve this discrepancy, we combined optogenetic 5-HT stimulation with large-scale Neuropixel recordings across the forebrain of lightly anesthetized mice. We demonstrate that selective 5-HT release consistently induces DOWN states across the cortex and striatum, while leaving midbrain dynamics largely unaffected. Using a multi-area computational model constrained by biologically plausible connectivity, we show that these transitions are not solely dependent on direct DRN innervation. Instead, 5-HT-induced state transitions propagate to remote cortical regions via network-level interactions.

Introduction

Serotonin (5-HT) is a neuromodulator which plays a crucial role in many brain functions and modulates neural activity in various ways [1]. One of the activity patterns in the brain which is modulated by serotonin are slow oscillations, with alternating states of high and low activity, also referred to as UP and DOWN states. Slow oscillations appear during anesthesia and slow-wave (non-REM) sleep and are important for learning, memory consolidation and plasticity [2–5], they are thought to be the default emergent activity of neural networks [6]. Although it is clear that serotonin affects slow oscillations, different studies arrive at conflicting conclusions as to the directionality of its effect.

It was shown that increasing tonic levels of serotonin by administering MDMA suppresses slow oscillations in anesthetized mice. Furthermore, bath application of 5-HT to acute brain slices completely abolishes slow oscillations [7]. Besides these slow tonic increases in serotonin, also transient pulses of 5-HT suppress neural activity across the brain, as measured with fMRI [8, 9]. Although these studies suggest that serotonin induces a DOWN state, they lack the temporal precision to directly establish this. Another piece of evidence which supports the notion that serotonin should induce a DOWN state is that 5-HT stimulation suppresses hippocampal sharp wave ripples [10, 11]. Since sharp wave ripples occur during UP states it is likely that serotonin transitions the neural network into a DOWN state. This is, however, only indirect evidence. In contrast to the previous findings, one study showed that electrical stimulation of the dorsal raphe nucleus (DRN) in rats induces an UP state in frontal cortex [12]. To our knowledge, this is the only study which investigated the direct effect of a transient increase of serotonin on UP-DOWN dynamics with high temporal resolution. Their findings, however, are in direct conflict with a body

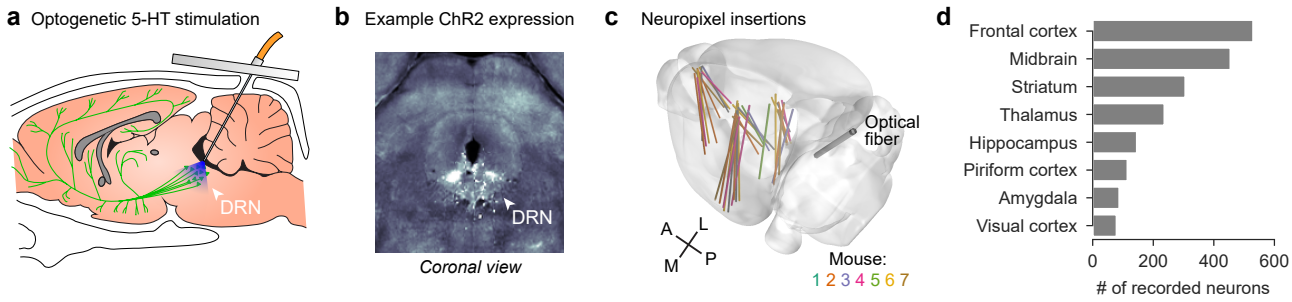


Figure 1. Acute Neuropixel recordings while optogenetically stimulating serotonin release. **(a)** Saggital cartoon drawing of serotonergic neurons in the DRN projecting throughout the brain (green). An optical fiber is positioned directly above the DRN through the cerebellum through which the 5-HT neurons can be activated with blue light. **(b)** A coronal histological section of the DRN showing expression of channelrhodopsin (ChR2). **(c)** Neuropixel insertion trajectories, aligned to the Allen Brain reference atlas, across all seven mice. Seven insertions were made per mouse over a period of four days, typically two at the same time. Color of each insertion indicates mouse identity. **(d)** The total number of recorded neurons across all recording sessions split out by brain region.

of literature which indicates that serotonin suppresses neural activity and transitions the network into a DOWN state. Serotonin's effect on UP-DOWN state dynamics under anesthesia is therefore still a topic for debate.

To shed light on this issue, we stimulated serotonin by optogenetically activating DRN neurons in the anesthetized mouse while recording single cell spiking activity across the brain with Neuropixel probes. We found that 5-HT stimulation rapidly induced a DOWN state in cortex, amygdala, striatum, hippocampus, thalamus, but not the midbrain. To investigate the underlying mechanism of this effect we modelled UP-DOWN state dynamics with a firing rate model with realistic inter-area connectivity which replicates the experimental data [13, 14].

Results

We performed acute Neuropixel recordings across the mouse brain while the animal was under light anesthesia. Over four consecutive recording days, seven Neuropixel insertions were made targeting frontal cortex, visual cortex, piriform cortex, amygdala, striatum, hippocampus, thalamus, and the midbrain. Serotonin was stimulated by activating serotonergic neurons in the DRN optogenetically during Neuropixel recordings in the forebrain (Fig. 1). Serial-sectioning two-photon microscopy was used to reconstruct the entire brain volume post-mortem. The reconstructed brain volume was used to determine channelrhodopsin expression in the DRN and reconstruct the Neuropixel insertion trajectories across the brain (Fig. S1). Out of seven mice, five showed good expression of channelrhodopsin in the DRN and were included in subsequent analysis, the two lacking expression, most likely due to mistargeted viral injections, were used as controls.

Cortical UP-DOWN states were clearly visible under anesthesia as periods of 1-2s of neural activity interspersed with periods of silence (Fig. 2a). Based on visual inspection of the spike raster plots, 5-HT stimulation appeared to coincide with a DOWN state. To investigate this, first we asked what the effect of serotonin stimulation was on the activity of single neurons. Peri-stimulus time histograms of single neurons, conditioned on the start of the 5-HT stimulation, consistently showed that single neuron spiking activity was decreased (example neuron in Fig. 1b). In all recorded brain regions we found a large fraction of neurons that were significantly modulated by serotonin stimulation (30-60%; ZETA-test [15], $p < 0.05$; Fig. 2c). A large majority of single neurons were suppressed by 5-HT stimulation as quantified by their modulation index, which was predominantly < 0. The mean modulation index over neurons per brain region was negative indicating that all recorded regions were suppressed by serotonin stimulation (Fig. 2d).

To directly investigate whether serotonin stimulation induced a DOWN state we quantified UP and DOWN states with a two-state Hidden Markov Model (HMM) which was fit to spike rates conditioned to the start of the 5-HT stimulation (Fig. 2e,f). We found that serotonin stimulation

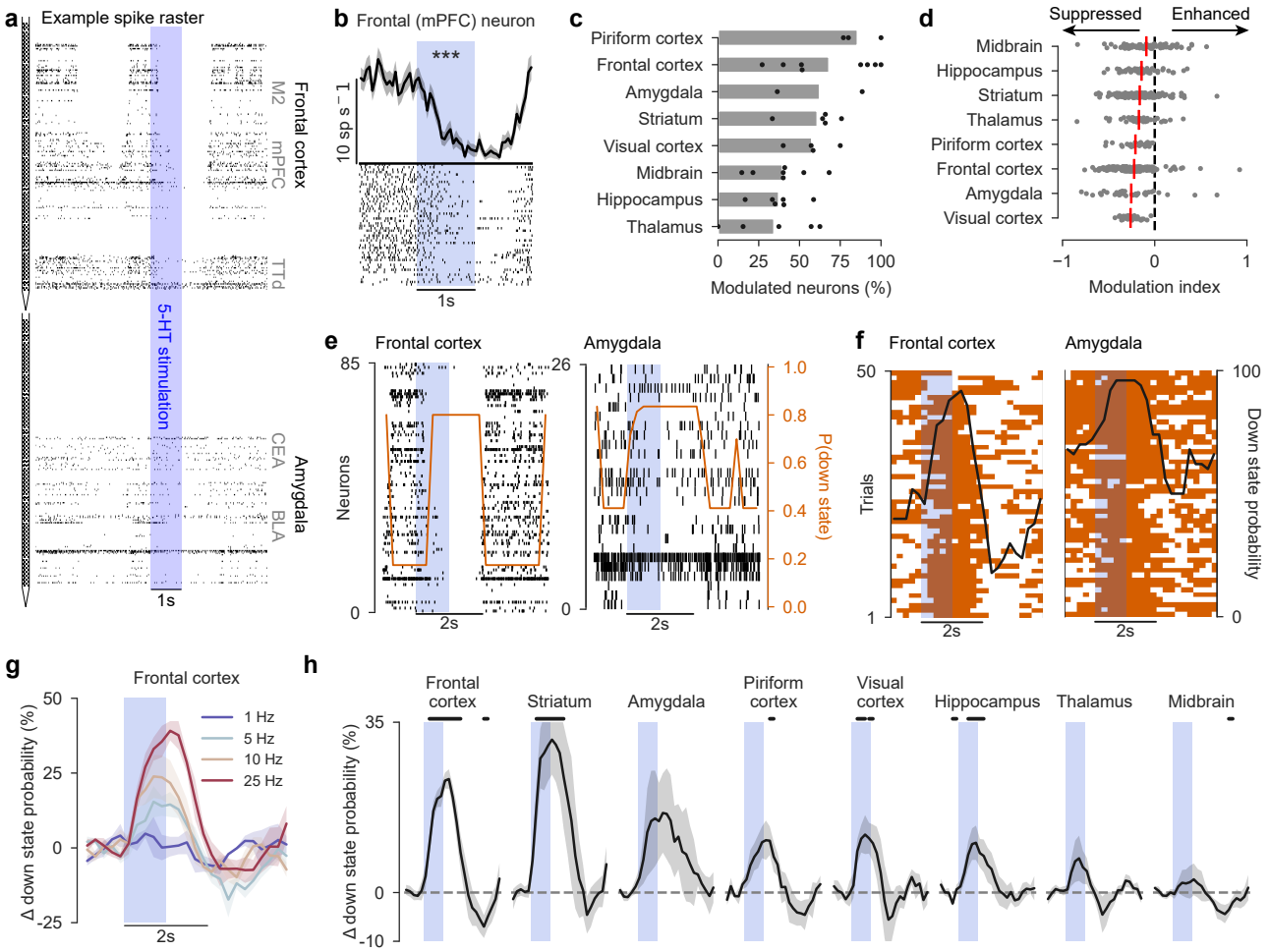


Figure 2. Serotonin induces a DOWN state under anesthesia. **(a)** An example spike raster plot of 10s of spiking activity recorded on two Neuropixel probes simultaneously. Recorded brain regions are indicated on the right of the raster at their approximate depths. The time of the optogenetic 5-HT stimulation is indicated by the blue bar (25 Hz, 10 ms pulses). **(b)** Peri-stimulus time histogram of an example neuron from the frontal cortex. The bottom shows the spike raster in which each row is a repetition of 1s 25 Hz stimulation. The top shows the mean spiking rate binned in 50 ms bins with a 25 ms Gaussian smoothing kernel. *** $p < 0.001$, ZETA-test. **(c)** Percentage of significantly modulated neurons (ZETA-test, $p < 0.05$) per brain region, calculated per recording session. Black dots are sessions, grey bars means over sessions. **(d)** Modulation index per neuron, negative values indicate the neuron is inhibited by 5-HT and positive values indicate excitation. Each grey dot is a single neuron, vertical red bars are the mean over neurons per brain region. **(e)** An example peri-stimulus raster plots of spike times (black ticks, each row is a neuron), aligned to the onset of 5-HT stimulation (blue vertical bar). The orange line indicates the posterior probability of a DOWN state as determined by a two-state HMM. On the left an example from the frontal cortex is shown and on the right an example from the amygdala. **(f)** A peri-stimulus heatmap over trials of when the HMM determined there most likely was a DOWN state (orange). The black line is the mean posterior probability of a DOWN state over trials per time bin. **(g)** A peri-stimulus plot of the change in DOWN state probability compared to baseline (-1 to 0s) averaged over mice (n = 5, mean ± sem). Plotted separately, indicated by color, are different stimulation frequencies. **(h)** Peri-stimulus plots for all recorded high-level brain regions of the change in DOWN state probability over baseline for 25 Hz serotonin stimulation. Black horizontal bars above the plot indicate periods of statistical significance (t-test versus 0, $p < 0.05$).

strongly and rapidly induced a DOWN state in the frontal cortex, this effect scaled with the stimulation frequency of the optogenetic stimulation (Fig. 2g). Performing the analysis across all recorded brain regions we found that serotonin strongly induced a DOWN state in the frontal cortex and striatum, to a lesser extend in the amygdala, visual cortex, hippocampus, and thalamus, and not at all in the midbrain (Fig. 2h). Importantly, these results could not be explained by neural responses to the optogenetic light delivery alone (Fig. S2).

We next investigated the physiological factors which contributed to the magnitude of the 5-HT

induced suppression of neural activity. A previous study optogenetic fMRI study reported no correlation with the projection density from the DRN with the 5-HT elicited drop in BOLD response but a significant correlation with 5-HT1f, 5-HT2a and 5-HT2c receptor expression [9]. We tried to replicate these findings by obtaining the DRN projection strength and receptor expression profiles for various serotonergic receptors from the Allen Brain Atlas and correlated them with the magnitude of the suppression of spiking activity elicited by serotonin stimulation at a region-by-region basis. Similarly to the fMRI study, we found no significant correlation with the DRN projection strength (Fig S3a). Out of all the serotonergic receptors expression profiles we were able to obtain from the Allen Brain Atlas we found significant correlations with 5-HT1f and 5-HT2a (but not 5-HT2c) expression such that higher levels of receptor expression were associated with stronger suppression of neural activity in that region (Fig S3b-d).

We aimed to shed further light on the mechanisms underlying serotonergic suppression of neural activity and the induction of DOWN states across the brain. To this end we implemented a dynamic bistable firing rate model to investigate the mechanism by which serotonin stimulation results in the induction of DOWN states. The model simulates a simplified network of the mouse brain, following biological realistic inter-area connectivity. Each brain area was modelled using a local excitatory-inhibitory (EI) circuit consisting of one E, one I population, and a mechanism controlling for adaptation (A) of the excitatory population [14] (Fig. 3a). To introduce biologically plausible connectivity between the modelled brain regions, the projection strength between them was pulled from the Mouse Connectivity Atlas of the Allen Brain Atlas, together with the projection strength of the dorsal raphe nucleus to these regions (Fig 3b. The model was able to reproduce slow-wave oscillations which mimicked closely the UP and DOWN states observed in the neural data (Fig. 3c). The frequency of modelled UP and DOWN states was around 0.5-1 Hz which closely matched the frequency observed in the brain.

Serotonin stimulation was modelled by targeting the E population of each brain region with an inhibitory current which was proportional to the projection strength from the DRN to that region. To investigate the effect of the modelled serotonin stimulation on UP-DOWN state dynamics a two-state HMM was fitted to the model output. We found that, in the model, serotonin stimulation induced a DOWN state across brain regions in a qualitatively similar matter compared to the actual neural data (Fig. 3d). One notable exception was the hippocampus which showed a weak DOWN state induction in the neural data but none at all in the model.

It was chosen to model serotonin stimulation as an inhibitory current targeting the E population ($E-$), as this increases the threshold for the E population to fire, making it likely to induce a transition from the UP to the DOWN state. Intuitively, targeting the I population with an excitatory current ($I+$) should result in a similar outcome. The two other possibilities are: targeting the E population with an excitatory current ($E+$), and the I population with an inhibitory current ($I-$). Running the model with all four scenarios revealed that only the $E-$ scenario resulted in a transition to a DOWN state (Fig. 3e). Contrary to our expectation, the $I+$ scenario did not in the induction of a DOWN state.

We next investigated the importance of biologically realistic connectivity in the model by systematically varying the coupling strength G. In an completely uncoupled network ($G = 0$), the probability of a DOWN state in each region during simulated 5-HT stimulation was strongly correlated with the projection from the DRN (Pearson correlation, r=0.83, p=0.01; Fig. 3f, left). At a high level of global coupling ($G = 3$) there was still a significant correlation, albeit a bit weaker (r=0.73, p=0.04; Fig. 3f, right).

We observed in Fig. 3f that some regions showed a large change in the probability of a DOWN state between the uncoupled and connected model whereas other regions did not show any difference. This was quantified by taking the Δ down state between the $G = 0$ and $G = 3$ models for each brain region (Fig. 3g). This revealed that the visual cortex showed the largest change in down state probability compared to the other brain regions by far (Fig. 3h). Notably, it was not the case that any region receiving weak DRN input showed a large change in down state probability since the Δ down state was not correlated with DRN input strength (data not shown). This means that the 5-HT induced DOWN state in visual cortex is not due to a direct projection from the DRN but instead because the visual cortex starts following the UP-DOWN state dynamics of the rest of the cortex through inter-area communication.

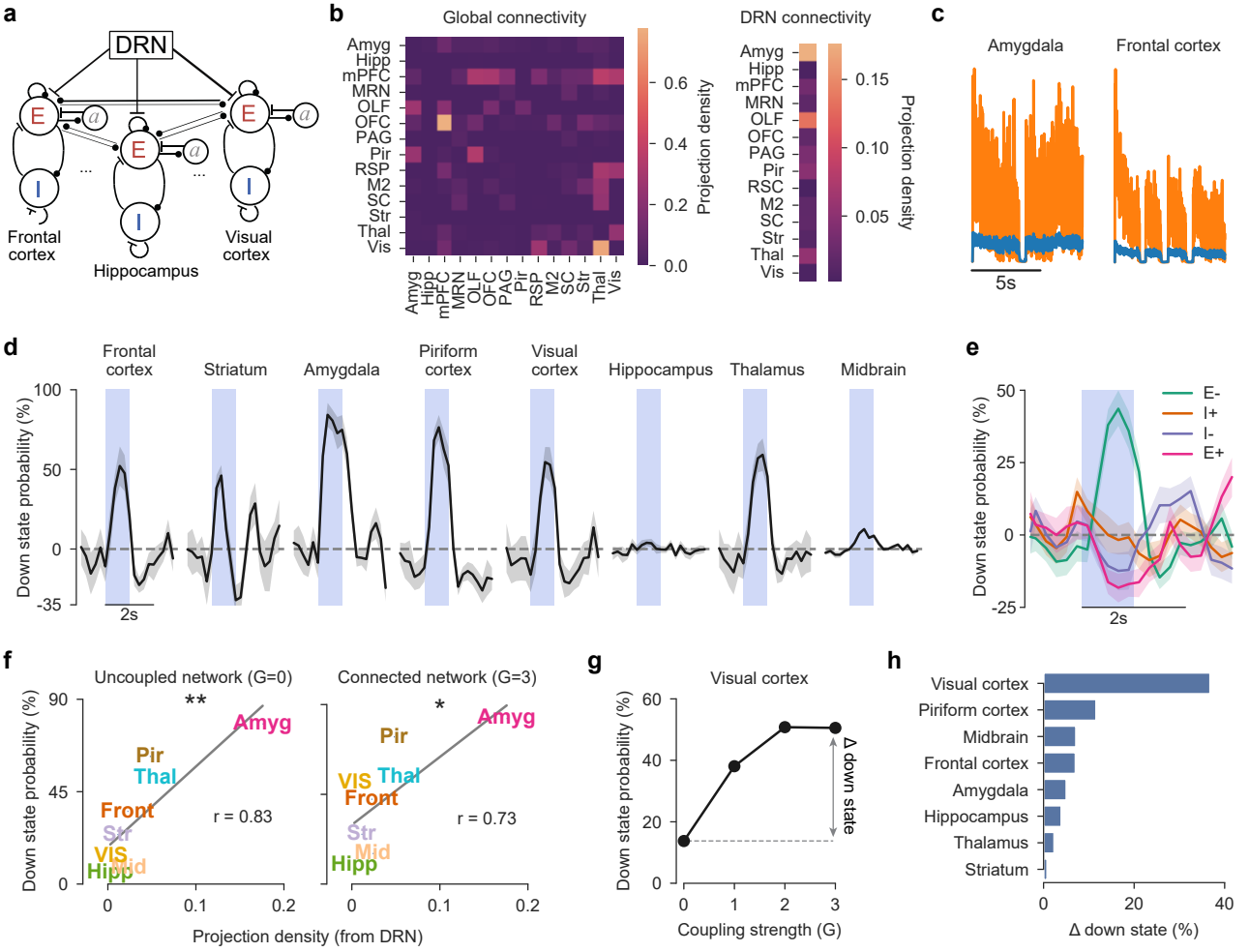


Figure 3. Simulation of UP-DOWN state dynamics by a bistable firing rate model. **(a)** A global circuit model of interconnected brain regions which receive input from the dorsal raphe nucleus (DRN). **(b)** Connectivity matrix for the global circuit and DRN projections. **(c)** The modelled activity recapitulated UP-DOWN state dynamics in brain regions which also show these dynamics in the brain; e.g. the frontal cortex. **(d)** Modelled serotonin stimulation induced a DOWN state in several brain regions. The change in DOWN state probability versus baseline (-1 to 0s) is plotted per brain region. **(e)** DOWN states were induced by inhibiting the excitatory population (E-), but not by exciting the inhibitory population (I+), inhibiting the inhibitory population (I-), or exciting the excitatory population (E+). The change in DOWN state probability is plotted as in panel d, averaged over regions. **(f)** In an uncoupled network (coupling strength G=0, left panel), the DRN projection density to a given brain region determines the strength of the effect of the modulation by serotonin. Brain regions are colored words. Amyg: amygdala, Pir: piriform cortex, Thal: thalamus, Front: frontal cortex, Str: striatum, VIS: visual cortex, Mid: midbrain, Hipp: hippocampus. ** p=0.01, r=0.83, Pearson correlation. In a network with strong realistic connectivity between brain regions (G=3, right panel) the DRN projection the correlation is less strong. * p=0.04, r=0.73, Pearson correlation. **(g)** The probability of a DOWN state, as defined by the maximum during 5-HT stimulation (0-1s), plotted for increasing values of inter-area coupling strength G in the network. Here only visual cortex is plotted. The effect of coupling strength on DOWN state probability is defined as the absolute difference between $G = 0$ and $G = 3$. **(h)** The Δ down state, as defined in panel g, per brain region.

Discussion

We showed that selective activation of serotonergic neurons in the DRN under light anesthesia resulted in an inhibition of neural activity across the mouse brain. This is in stark contrast with the effect of serotonin stimulation in the awake condition where, in a previous study, we found that serotonin had a bidirectional effect on spiking activity [11]. Our aim here was to resolve a discrepancy in the literature; optogenetic fMRI showed a brain-wide decrease in neural activity when stimulating 5-HT under anesthesia [8, 9] while electrical stimulation of the DRN under

anesthesia resulted in increased activity and the induction of a cortical UP state [12]. Our results are in line with the fMRI studies, we show here conclusively that serotonin stimulation results in a brain wide reduction of single-neuron spiking activity accompanied by the induction of a DOWN state across the cortex and striatum. Computational modelling further revealed that cortical regions which lack direct DRN input, like the visual cortex, are still transitioned into a DOWN state due to network-level interactions.

The most likely explanation for the contradictory results of [12] is that electrical stimulation of the DRN is non-selective and activates glutamatergic as well as serotonergic neurons. The glutamatergic drive is apparently much stronger compared to the serotonergic one and dominates the down-stream effect resulting in the induction of an UP state instead of a DOWN state. Another possibility, albeit less likely, is that the study of [12] was performed in rats whereas the other studies, including this one, were performed in mice. To our knowledge, however, there is no reason for the effect of serotonin to be opposite in rats versus mice.

Slow oscillations, or UP-DOWN states, are observed in the cortex, thalamus and striatum during non-REM sleep. Contrary to REM sleep, during which serotonin is tonically low, serotonin levels in cortex show phasic peaks during non-REM sleep [16]. This suggests that, during non-REM sleep, serotonin serves as a state switch between periods of UP-DOWN state dynamics and periods of overall suppression of neural activity. Whether isoflurane-induced UP-DOWN state dynamics are mechanistically equivalent to those observed during natural slow-wave sleep is debatable. However, a direct comparison of UP-DOWN state characteristics between non-REM sleep and light isoflurane anesthesia showed that they are fairly similar [17].

The amygdala and hippocampus do not normally show UP-DOWN state dynamics, however, they did show increased DOWN state probability in response to 5-HT stimulation. This is likely not the system transitioning into a DOWN state, since they do not naturally occur, but simply a suppression of neural activity which is picked up by the two-state HMM as a DOWN state. This suggests that the induced DOWN state in the amygdala and hippocampus are qualitatively different from those that are induced in regions which do naturally exhibit UP-DOWN dynamics, such as the cortex, thalamus, and striatum. Follow-up experiments are necessary to say anything conclusive in this respect.

Besides serotonin, some serotonergic neurons in the DRN co-release glutamate. Although the percentage is relatively low among serotonergic neurons projecting to subcortical areas ($\tilde{1}5\%$), it is much higher in serotonergic neurons projecting to cortex ($\tilde{6}0\%$; [18]). The co-release of glutamate should result in excitation of down stream neuronal populations, which is something we putatively observed in the awake condition [11]. During light anesthesia, however, the excitatory glutamatergic drive is masked by the dominant suppressive effect of serotonin release. Targeted pharmacological studies are required to determine exactly which biophysical mechanism causes the strong inhibitory drive under anesthesia. Our correlative results suggest that the 5-HT1f and 5-HT2a receptors are involved.

A surprising finding was that in the electrophysiological data there was no correlation between the strength of the 5-HT induced suppression and the projection strength of the DRN to that region (Fig. S3a). In the awake condition we found the same result [11]. Our computational model provided a possible explanation; when increasing the inter-area connectivity in the model the correlation with DRN projection strength went down (Fig. 3f). In the computational model, however, the correlation with DRN input never completely disappeared. This was likely due to the fact that the model was a simplified version of reality which lacked physiological detail, such as serotonergic receptor expression patterns and volume transmission of serotonin across region boundaries. Regardless, the model showed that a subset of brain regions, lacking direct DRN input, are still transitioned into a DOWN state through network-level synchronization of neural activity.

Data and code availability

All these data are open-source available through the Open Neurophysiology Environment (ONE) protocol (https://int-brain-lab.github.io/ONE/). All the code used to generate the figures in this manuscript are on Github: https://github.com/rosagross/SerotoninAnaesthesiaModel/.

Material and methods

Animals

Seven SERT-Cre C57Bl/6 mice (four males and three females) were used for the purpose of this study. All experimental procedures were approved by the Portuguese Veterinary General Board.

Surgical procedures

Mice underwent two surgeries: (1) injection of the channelrhodopsin viral vector in the dorsal raphe nucleus and implantation of the optical fiber & headbar, and (2) the creation of three small craniotomies in the skull. Both these surgical procedures were performed as previously described in [11]. A minimum of two weeks was allowed for the virus to express and the mouse to recover after the first surgery. The second surgery was performed $\tilde{1}2$ hours preceding the first Neuropixel recording.

Neuropixel recordings

Acute anesthetized Neurpixel recordings were performed on four consecutive days. Mice were anesthetized with an induction dose of 3% isoflurane after which they were headfixed in the recording rig (hardware described in detail in [11]). A maintenance level of isoflurane of 0.8-1.5% was administered through a mask which was placed over the nose of the mouse. The tip of each Neuropixel probe was lowered in drop of Di-I with the micromanipulator prior to insertion. One or two Neuropixel probes, depending on the recording day, were inserted in the following coordinates:

Session	ML (μm)	AP (μm)	Depth (μm)	Angle ($^\circ$)	Directionality
1	-1090	2000	3800	15	medial
1	-2184	-2000	6800	10	lateral
2	-946	-3300	3800	10	medial
2	-1227	2000	4590	10	lateral
3	-2240	-2000	4000	15	medial
3	-850	-3460	5270	10	lateral
4	-2250	-1800	5500	17	lateral

Throughout the recording session, the isoflurane level was calibrated such that the mouse would not wake up but UP and DOWN states were visible in the electrophysiological recordings. Once mice were stable under light anesthesia, serotonergic neurons in the dorsal raphe nucleus were optogenetically stimulated with 1s pulse trains of blue light (465 mm) administered through the optical fiber. The light power was 5 mW, as measured at the tip of the optical fiber. Light pulses were 10 ms long and with a frequency of 1, 5, 10, or 25 Hz. Different frequencies were administered in a randomized order and repeated each 50 times. The inter-stimulus interval was drawn from a truncated exponential distribution with a factor of 6s, a minimum of 5s, and a maximum of 15s. The whole stimulation protocol lasted around 30 minutes, during this time the body temperature of the mice was kept from falling using hand warmer packages (electrical heating pads produced large artifacts in the electrophysiological recordings). After the protocol, the Neuropixels were retracted, the skull covered, and the animal was placed in a cage positioned on a heating pad to wake up.

Processing of electrophysiological data

Pre-processing and spike sorting of raw electrophysiological data was performed with the fully automated pipeline developed by the International Brain Laboratory (IBL) [19]. We only included neurons that passed all the IBL neuron-level quality control criteria. Additionally we excluded neurons with firing rates < 0.1 spks/s over the whole recording.

Modulation index

For each neuron a modulation index was calculated which quantifies the strength and directionality of the modulatory effect of serotonin. A 500 ms baseline window was defined directly before

stimulus onset, and a 500 ms stimulus window was defined at 300 - 800 ms after stimulus onset. The modulation index MI was defined as the area under the ROC curve $auROC$ between the spike counts in the baseline window and the stimulus window, which was scaled between -1 and 1 according to Equation 1.

$$MI = 2(auROC - 0.5) \tag{1}$$

Firing rate model

We implemented a bistable firing rate model, consisting of multiple local circuits which were all-to-all connected to form a global network. The local circuits represented the relevant brain areas from the experimental recordings. Serotonin stimulation was modelled by inducing a current to all regions, originating from the DRN. The connectivity was realistically modelled by using connectivity data from the Allen Brain Atlas which defined the connectivity between the regions in the model as well as the projection density from the DRN to the different regions.

The local circuit model was based on the implementation by Jercog et al. [14], including one excitatory population E and one inhibitory population I. It included an adaptation module a, simulating synaptic fatigue in pyramidal cells, as well as background noise which was introduced to the E and I population. Both populations projected to each other, as well as to themselves through self-coupling. The following equations describe the change of the populations E and I firing rate r (see Equation 2 and 3), as well as the strength of adaptation a (see Equation 4) over time, where τ_E, τ_I and τ_a are time constants:

$$\tau_E \frac{dr_E}{dt} = -r_E(t) + \varphi_E(J_{EE}r_E(t) - J_{EI}r_I(t) - a(t) + \sigma\xi_E(t)) \tag{2}$$

$$\tau_I \frac{dr_I}{dt} = -r_I(t) + \varphi_I(J_{IE}r_E(t) - J_{II}r_I(t) + \sigma\xi_I(t)) \tag{3}$$

$$\tau_a \frac{da}{dt} = -a(t) + \beta r_E(t) \tag{4}$$

The frequency of UP and DOWN state changes was defined by $\frac{1}{\tau_a}$, so the original value of $\tau_a = 500ms$ generated states in the frequency of around 2Hz. Analyses of the experimental data in the mouse revealed that the frequency of state change is 0.5-1Hz, therefore a value of $\tau_a = 1500ms$ was chosen and the other time constants were scaled accordingly, so that the final implementation yielded $\tau_E = 30ms$ and $\tau_I = 6ms$. The variable J defined the local coupling strength for the E and I population, which was $J_EI = 5$, $J_EE = 1$, $J_EE = 10$, and $J_EE = 0.5$ (the first letter indicating the target population). To model the noise of the system, which was introduced as a background noise to both populations, two Ornstein-Uhlenbeck processes were implemented, with a mean of zero and a standard deviation of $\sigma = 3.5$. The functions φ_E and φ_I were the transfer functions, which determined the reaction of the population to incoming currents, through self coupling, projections coming from other populations, or/and background noise. The equation is a threshold-linear function, with a threshold that defines the start of the linear part of the equation. If the input did not reach the threshold, the activation of the regarding population was set to zero.

$$\varphi_X(x) = g_X[x - \theta_X]_+ \qquad , X = \{E, I\} \tag{5}$$

The threshold that has to be reached until the population reacts to incoming input is described by θ. For brain regions that showed UP and DOWN states $\theta_E = 6$ and $\theta_I = 25$ were used as thresholds. In case the input reached the threshold level, the slope g determined the output. g controlled how sensitive the population reacts to input; a higher g will lead to a stronger output than a lower g. The slope g was fine-tuned for each brain region so that the average firing rate of the E and I population is slightly below the value measured in the mouse brain, since inter-area connectivity further increases the baseline firing rate. UP and DOWN state characteristics were also matched to the experimental data on a region-by-region level; some brain regions, like the cortex, showed strong UP-DOWN dynamics *in vivo* whereas other regions, like the midbrain, did not. To mimic this in the

model, the adaptation strength β and the threshold of the E population θ_E were fine-tuned per brain region. For regions which showed strong UP-DOWN dynamics $\beta = 6$ and $\theta_E = 4$ was used, regions without UP-DOWN dynamics were modelled at $\beta = 1$ and $\theta_E = 5$, and intermediate regions at $\beta = 3$ and $\theta_E = 5$.

Modelling inter-area connectivity

Realistic connectivity was introduced to the model using data from the Mouse Connectivity Atlas (Allen Brain Institute). Brain regions which were targeted by the Neuropixel recordings were defined as regions as interest. For each region, we obtained the projection density values to all other regions. Results from all experiments per region were averaged. Since some experiments did not show any projection values, a threshold of a projection density of 0.005 was applied for the experiment to be included. As a next step, the obtained values of the fine grained areas were grouped together into higher level brain areas that each represent one component of the network (see Figure **??**). Besides the connectivity values between the areas within the network, the anatomical projections originating from the DRN were set according to tracing data from the Allen Brain Institute (see Figure **??**). The connectivity data was introduced to the global network model, by taking the projection density values W^{xy} (projection to area x from area y) and multiplying them with the coupling strength G between the inter-area projections. The input into population E of area x can therefore be defined as I_E^x from projections that originate from all other areas y (see Equation 6).

$$I_E^x = G \sum_y W^{xy} \tag{6}$$

Modelling 5-HT stimulation

The modulatory effect of serotonin stimulation was modelled by a current from the DRN projecting to the local circuits in the model. Similar to the connectivity matrix for the inter-area long-range projections, the projections of the DRN are weighted depending on the projection density. Considering that the DRN projection densities to all areas x are stored in a vector $\mathbf{D}$ the input current I_{DRN} is defined as the following:

$$I_{DRN}^x(t) = S\mathbf{D}_x u(t) \tag{7}$$

The temporal dynamics of the release of serotonin in the model was matched to the temporal profile of the modulatory effect of 5-HT on neural activity (data not shown). The trajectory of 5-HT modulation peaked after 0.5 seconds and decayed it returns to the original state after 3 seconds. The time course of this trajectory, here defined as $u(t)$, was multiplied with the parameter for serotonin stimulation strength S. The stimulation was given for 1 second, with an inter-stimulus interval similar to the experimental protocol. The effect of serotonin stimulation on the system was initially modelled by introducing an inhibitory current to the E population (subtracting I_{DRN} from r_E), this was later extended to other mechanisms.

Hidden Markov Model (HMM)

In both the electrophysiological and simulated data, UP and DOWN states were quantified by fitting an HMM using the `ssm` package [20]. In the case of the electrophysiological data, spike times were binned in 200 ms bins in seven second windows (2s pre, 5s post) surrounding 5-HT stimulation onsets. An HMM with two states and Poisson observations was fit to the binned spike rates. In the case of the model, an HMM with two states and Gaussian observations was fit to the modelled firing rate time series of the I population. The state which showed the highest firing rate was determined to be the UP state. The output of the HMM was an inferred UP or DOWN state per time point and a corresponding posterior likelihood.

References

[1] P. Celada, M. Victoria Puig, and F. Artigas, Serotonin modulation of cortical neurons and networks, *Frontiers in Integrative Neuroscience* **7**, 25 (2013), ISSN 16625145.

[2] M. Steriade, D. A. McCormick, and T. J. Sejnowski, Thalamocortical Oscillations in the Sleeping and Aroused Brain, *Science* **262**, 679 (1993).

[3] M. Steriade, I. Timofeev, and F. Grenier, Natural Waking and Sleep States: A View From Inside Neocortical Neurons, *Journal of Neurophysiology* **85**, 1969 (2001), ISSN 0022-3077, 1522-1598.

[4] G. T. Neske, The Slow Oscillation in Cortical and Thalamic Networks: Mechanisms and Functions, *Frontiers in Neural Circuits* **9**, 88 (2015), ISSN 1662-5110.

[5] C. J. Wilson and P. M. Groves, Spontaneous firing patterns of identified spiny neurons in the rat neostriatum, *Brain Research* **220**, 67 (1981), ISSN 0006-8993.

[6] M. V. Sanchez-Vives, M. Massimini, and M. Mattia, Shaping the Default Activity Pattern of the Cortical Network, *Neuron* **94**, 993 (2017), ISSN 0896-6273.

[7] R. de Filippo, B. R. Rost, A. Stumpf, C. Cooper, J. J. Tukker, C. Harms, P. Beed, and D. Schmitz, Somatostatin interneurons activated by 5-HT2A receptor suppress slow oscillations in medial entorhinal cortex, *eLife* **10**, e66960 (2021), ISSN 2050-084X.

[8] H. T. Hamada, Y. Abe, N. Takata, M. Taira, K. F. Tanaka, and K. Doya, Optogenetic activation of dorsal raphe serotonin neurons induces a brain-wide response in reward network, *bioRxiv* 2022.08.07.503074 (2022).

[9] J. Grandjean, A. Corcoba, M. C. Kahn, A. L. Upton, E. S. Deneris, E. Seifritz, F. Helmchen, E. O. Mann, M. Rudin, and B. J. Saab, A brain-wide functional map of the serotonergic responses to acute stress and fluoxetine, *Nature Communications 2019 10:1* **10**, 1 (2019), ISSN 2041-1723.

[10] C. L. Kandilakis and C. Papatheodoropoulos, Opposite serotonergic modulation of sharp waves in the dorsal and ventral hippocampus, *Frontiers in Synaptic Neuroscience* **17** (2025), ISSN 1663-3563.

[11] G. T. Meijer, J. A. Catarino, L. Freitas-Silva, I. Laranjeira, I. B. Laboratory, and Z. F. Mainen, Serotonin drives choice-independent reconfiguration of distributed neural activity, *bioRxiv* 2025.08.01.668048 (2025), ISSN 2692-8205.

[12] M. V. Puig, A. Watakabe, M. Ushimaru, T. Yamamori, and Y. Kawaguchi, Serotonin Modulates Fast-Spiking Interneuron and Synchronous Activity in the Rat Prefrontal Cortex through 5-HT1A and 5-HT2A Receptors, *Journal of Neuroscience* **30**, 2211 (2010), ISSN 0270-6474.

[13] J. F. Mejias, H. J. Kappen, and J. J. Torres, Irregular Dynamics in Up and Down Cortical States, *PLOS ONE* **5**, e13651 (2010), ISSN 1932-6203.

[14] D. Jercog, A. Roxin, P. Barthó, A. Luczak, A. Compte, and J. De La Rocha, UP-DOWN cortical dynamics reflect state transitions in a bistable network, *eLife* **6** (2017), ISSN 2050084X.

[15] J. S. Montijn, K. Seignette, M. H. Howlett, J. L. Cazemier, M. Kamermans, C. N. Levelt, and J. A. Heimel, A parameter-free statistical test for neuronal responsiveness, *eLife* **10**, e71969 (2021), ISSN 2050-084X.

[16] F. Deng, J. Wan, G. Li, H. Dong, X. Xia, Y. Wang, X. Li, C. Zhuang, Y. Zheng, L. Liu, et al., Improved green and red GRAB sensors for monitoring spatiotemporal serotonin release in vivo, *Nature Methods* **21**, 692 (2024), ISSN 1548-7105.

[17] M. Torao-Angosto, A. Manasanch, M. Mattia, and M. V. Sanchez-Vives, Up and Down States During Slow Oscillations in Slow-Wave Sleep and Different Levels of Anesthesia, *Frontiers in Systems Neuroscience* **15** (2021), ISSN 1662-5137.

[18] J. Ren, D. Friedmann, J. Xiong, C. D. Liu, B. R. Ferguson, T. Weerakkody, K. E. DeLoach, C. Ran, A. Pun, Y. Sun, et al., Anatomically Defined and Functionally Distinct Dorsal Raphe Serotonin Sub-systems, *Cell* **175**, 472 (2018), ISSN 0092-8674.

[19] I. B. Laboratory, K. Banga, J. Boussard, G. Chapius, M. Faulkner, K. D. Harris, J. Huntenburg, C. Hurwitz, H. Dong Lee, L. Paninski, et al., Spike sorting pipeline for the International Brain Laboratory (2022).

[20] S. Linderman, B. Antin, D. Zoltowski, and J. Glaser, SSM: Bayesian learning and inference for state space models, `https://github.com/lindermanlab/ssm`.

Supplementary figures

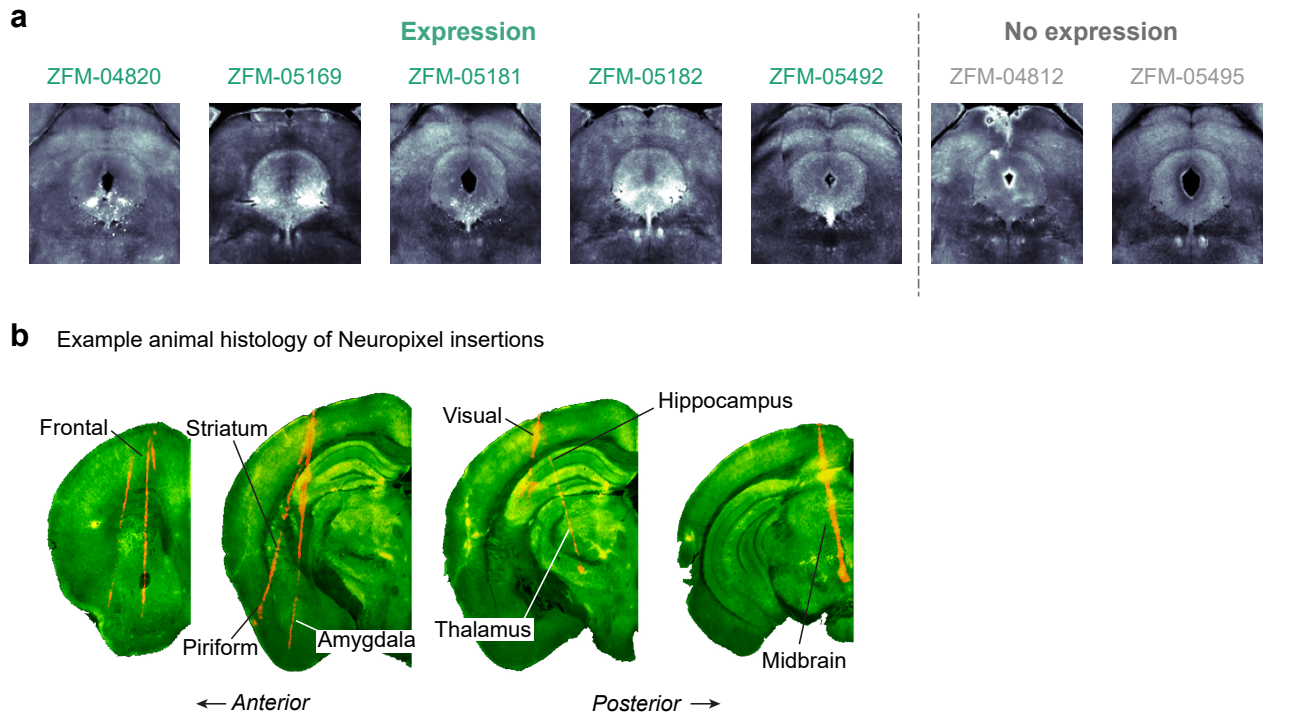


Figure S1. Histology. **(a)** Coronal slices of channelrhodopsin expression in the DRN for all seven SERT-cre mice. The first five mice, labeled in green, show expression of channelrhodopsin in the DRN as indicated by fluorescence directly underneath the ventricle. The last two mice, labeled in gray, do not show any expression due to a failed virus injection. **(b)** Fluorescent tracts of Neuropixel insertions. Neuropixels are dipped in Di-I resulting in a red fluorescent trace in the histology. Coronal slices at four different anterior-posterior positions are shown.

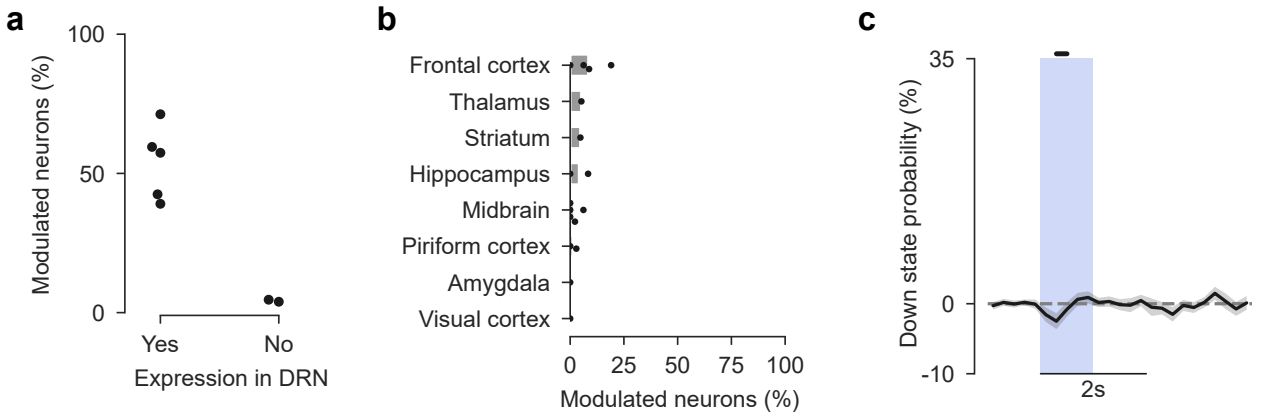


Figure S2. No light-evoked effect on neural activity in mice lacking channelrhodopsin expression. **(a)** The percentage of neurons which are significantly modulated by light delivery to the DRN. Each dot is a mouse, separated by whether that mouse expressed channelrhodopsin the DRN or not. **(b)** The percentage of light-modulated neurons in mice lacking channelrhodopsin expression, split out per brain region. Each dot is a recording session. **(c)** A very small decrease in DOWN state probability compared to baseline (-1 to 0s) was found in one time point during light delivery to the DRN (blue vertical bar). Data from all brain regions are pooled. Black horizontal bar above the plot indicates the significant time point (t-test versus 0, $p = 0.046$).

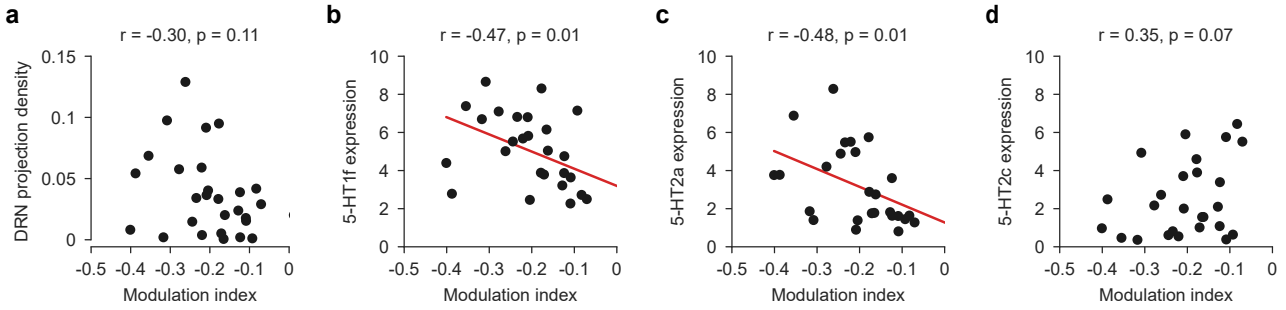


Figure S3. (a) No significant correlation between the projection strength from the dorsal raphe nucleus (DRN) with the magnitude of 5-HT induced suppression of neural activity. For each brain region, at a more granular delineation compared to the main figures, the projection density from the DRN was obtained from the Allen Brain Atlas. The modulation index of all neurons in each brain region was averaged. Each dot is a brain region. **(b)** There was a significant correlation for the expression strength of 5-HT1f receptors with 5-HT induced functional modulation at a region-by-region level. **(c)** Same as (b) for 5-HT2a receptor expression. **(d)** Same as (b) for 5-HT2c receptor expression. All stats done with a Pearson correlation.